

LE HAI TRUNG

VinUniversity, Hanoi, Vietnam

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EDUCATION

Bachelor of Mechanical Engineering - VinUniversity

9/2023 - Present

Current Cumulative GPA: 3.46 (Until 02/2026)

EXPERIENCE

Research Student in the Undergraduate Research Opportunities Program (UROP)

Biorobotics Laboratory of VinUniversity - Dr. Thai Mai Thanh

Project: Hybrid Soft Muscle-Tendon Glove for Hand Rehabilitation Post-Stroke

(Under Review for **Advanced Intelligent Systems**)

- Develop prototype and devices for post-stroke hand rehabilitation using soft artificial muscle rehabilitation and haptic devices.

Project: Integrating Servo-Controlled Soft Muscle Actuators with Tendon-Driven Mechanisms for a Teleoperated Humanoid Robotic Hand using Soft Filament Sensor

(Under Review for **Advanced Intelligent Systems**)

- Develop a servo-based control method integrated with tele-operation using angle-sensitive sensors for a remote control humanoid robot hand.

Tools & Frameworks: MATLAB, MATLAB Simulink, CAD Design, Soft Robotics

Project: 2D LiDAR-Based SLAM, 3D Environment Construction and Autonomous Navigation for PX4 UAV (On going)

- Develop and integrate a LiDAR-based SLAM system for real-time 3D mapping and localization of a UAV in ROS 2.
- Implement autonomous navigation and obstacle avoidance in PX4 SITL simulation using Gazebo.
- Evaluate mapping consistency, drift behavior, and loop-closure performance in multi-view simulated environments.

Internship: Intern 4 months at Phenikaa X - Team Drone

7/2025 - 10/2025

- Learn autopilot control using ROS2 OFBOARD control through an embedded computer combined with PX4 to program a precise landing UAV using Apriltag and computer vision.
- Obstacle avoidance for drones in GPS and Non-GPS environments using computer vision or combined proximity sensors.

Tools & Frameworks: PX4 Autopilot, Raspberry Pi, ROS2, CAD Design, Sensors

CERTIFICATES AND AWARDS

- IELTS: 7.5 (Taken August 11, 2022)
- Awarded 70% merit-based and 25% addition-based from **VinUniversity** in 2023.
- VinUniversity Dean's List Spring 2025.

SKILL SET

- CAD Design (Solidworks)
- Basics Simulation (ANSYS)
- MATLAB
- ROS2
- PX4 Autopilot, Ardupilot

INTERESTED FIELD

CAD Design - Autopilot system - Unmanned Vehicle - Soft Robotic - Mechatronics

REFERENCE

- Dr. Thai Mai Thanh 🌐
 - College of Engineering and Computer Science, VinUniversity, Vietnam.
 - Email: thanh.tm2@vinuni.edu.vn.
- VinUni BioRobotic Lab 🌐